MAS508-G Control Theory Lecture: Adaptive Dynamic Programming and LQR

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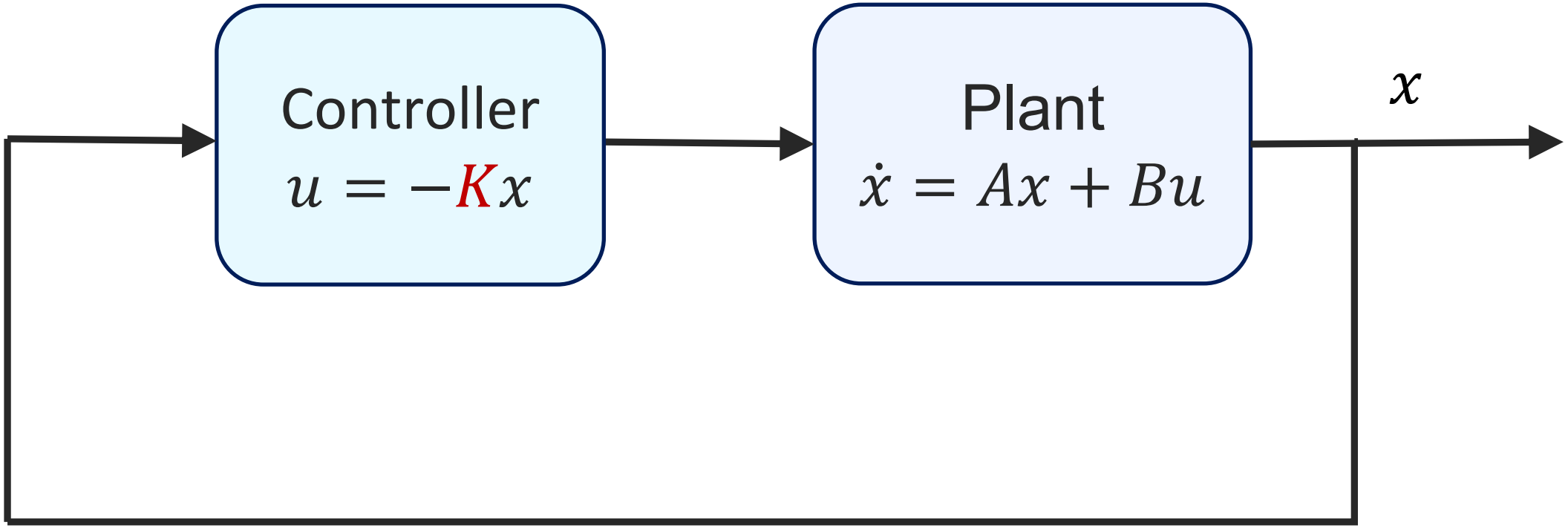
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Lecture. Adaptive Dynamic Programming and LQR

Linear Quadratic Regulator (LQR)

- Linear Quadratic Regulator (LQR) is an optimization method.
- It is used to **find the gain K** of a **state-feedback control.**



LQR

- For a system with the state-space model:

$$\dot{x} = Ax + Bu$$

- LQR finds a control law:

$$u = -Kx$$

- That minimizes the cost function

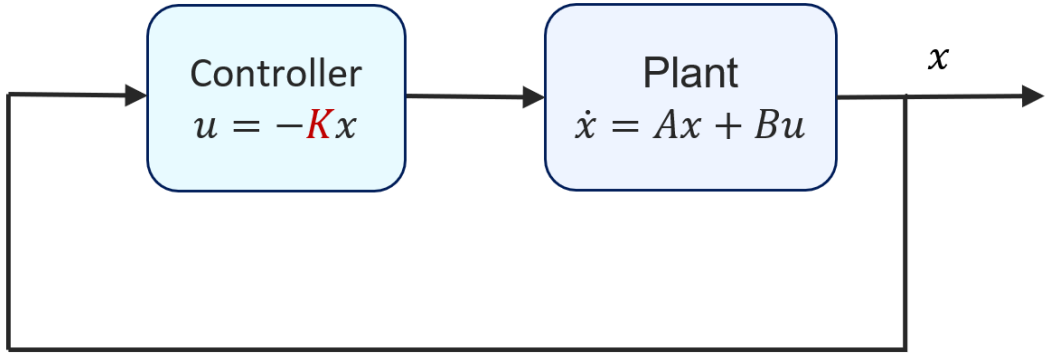
$$J = \int_0^\infty \underbrace{x(t)^T Qx(t)}_{\text{State cost with weighting matrix } Q} + \underbrace{u(t)^T Ru(t)}_{\text{Control cost with weighting matrix } R} dt$$

Performance Effort

- $Q \geq 0, R \geq 0$ are symmetric positive definite matrices.

LQR- Linear Quadratic Regulator

- LTI system $\dot{x} = Ax + Bu, x(0) = x_0$
- LQR is to **find the gain K** of a **state feedback control $u = -Kx$.**
- **To Minimize** $J = \int_0^\infty x(t)^T Qx(t) + u(t)^T Ru(t)dt$
- Find P that solves the Algebraic Riccati Equation. $A^T P + PA + Q - PBR^{-1}B^T P = 0$



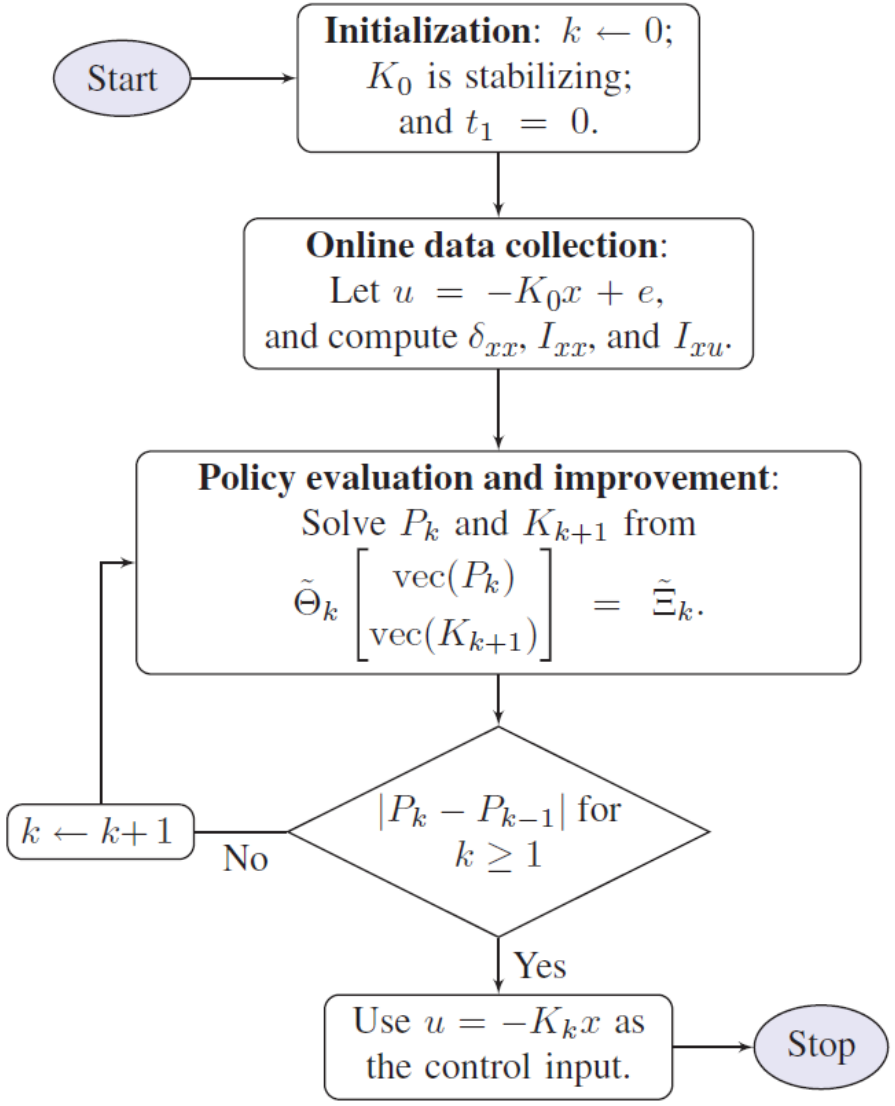
$$K = R^{-1}B^T P$$

Data-Driven LQR Design

- LTI system $\dot{x} = Ax + Bu, x(0) = x_0$
- When (A, B) is unknown, find a "data-driven" linear control policy
  $\boldsymbol{u = -Kx}$
- To minimize $J = \int_0^\infty x(t)^T Qx(t) + u(t)^T Ru(t)dt = x^T Px$

$Q \geq 0, R > 0$ are symmetric positive definite matrices.

ADP for LQR with unknown (A, B)



Off-line Off-policy ADP Policy Iteration

(1) *Initialization:*
Find K_0 such that $A - BK_0$ is Hurwitz. Let $k = 0$.

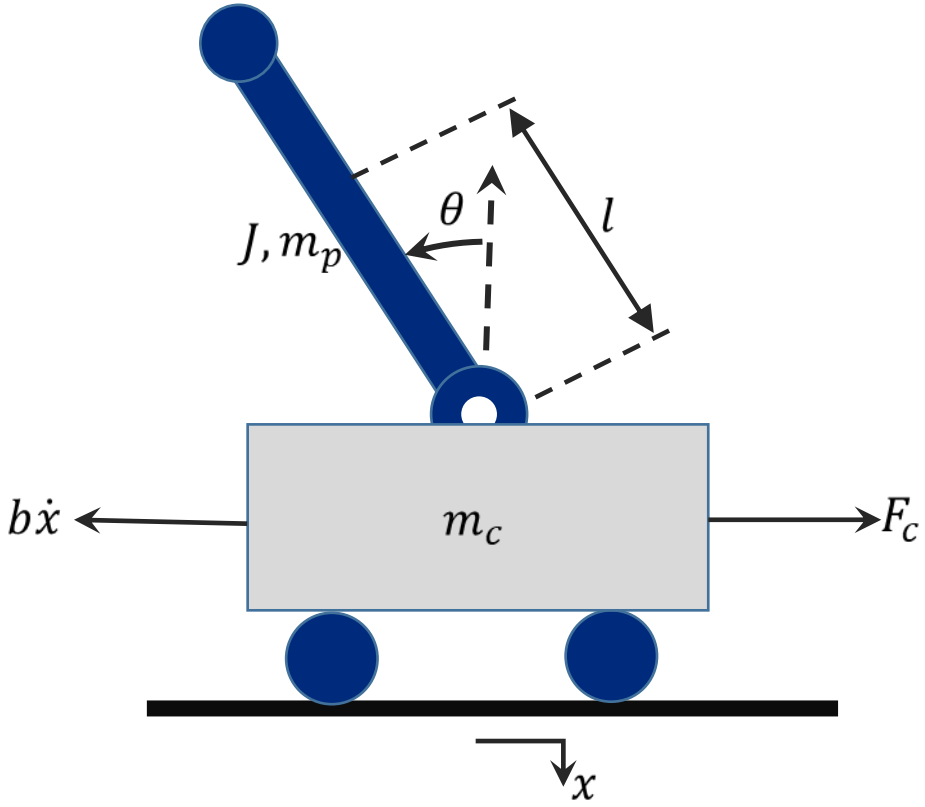
(2) *Online data collection:*
Apply $u_0 = -K_0 x + e$ as the control input from $t = t_1 = 0$, where K_0 is stabilizing and e is the exploration noise. Compute δ_{xx}, I_{xx}, and I_{xu} until the rank condition is satisfied. Let $k = 0$.

(3) *Policy evaluation and improvement:*
Solve P_k, with $P_k = P_k^T$, and K_{k+1} from $\tilde{\Theta}_k \begin{bmatrix} \text{vec}(P_k) \\ \text{vec}(K_{k+1}) \end{bmatrix} = \tilde{\Xi}_k$

(4) *Off-policy iteration:*
Let $k \leftarrow k+1$, and repeat Step (3), until $|P_k - P_{k-1}| \leq \epsilon$ for $k \geq 1$, where the constant $\epsilon > 0$ is a predefined small threshold.

(5) *Exploitation:*
Use $u = -K_k x$ as the approximate optimal control policy.

Example: Linear Inverted Pendulum




Segway

Linear equations of motion:

$$(m_c + m_p)\ddot{x} = F_c - b\,\dot{x} + m_p l\,\ddot{\theta}$$

$$(J + m_p l^2)\ddot{\theta} = m_p l\ddot{x} + m_p g l\theta$$

Define the state and control input:

$$x^T = [x \quad \dot{x} \quad \theta \quad \dot{\theta}] \qquad u = F_c$$

Linear state-space representation (after solving EOMs):

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -(J + m_p l^2)b/J_t & m_p^2 g l^2/J_t & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -m_p l b/J_t & m_p g l(m_c + m_p)/J_t & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ (J + m_p l^2)/J_t \\ 0 \\ (m_p l)/J_t \end{bmatrix} u$$

Where

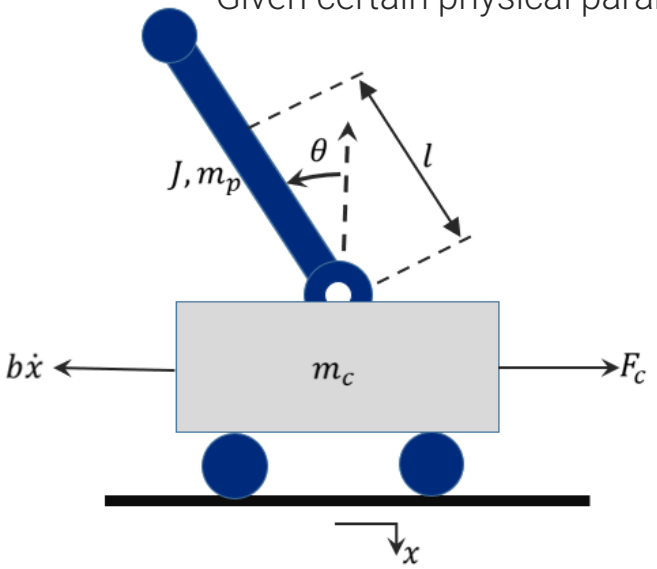
$$J_t = (J + m_p l^2)(m_c + m_p) - m_p^2 l^2$$

Example: Linear Inverted Pendulum

State-space model:

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -6.85 & 2.70 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -16.1 & 29.3 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 50.7 \\ 0 \\ 27.8 \end{bmatrix} u$$

Given certain physical parameters.



Controllability matrix:

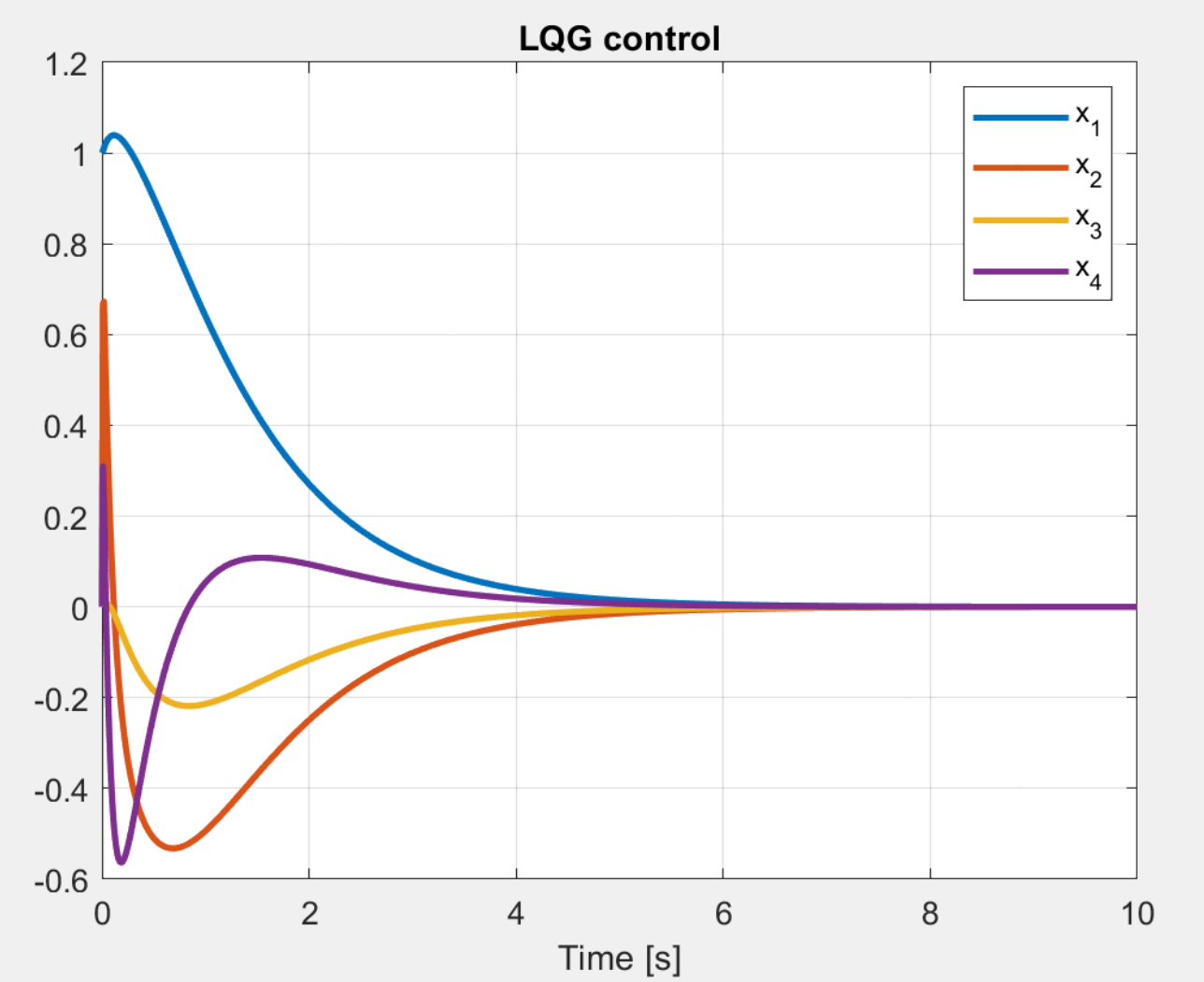
$$C_{ab} = [B \quad AB \quad A^2B \quad A^3B]$$

Rank of controllability matrix:

$$rank(C_{ab}) = 4$$

System is controllable.

LQR with known (A, B)

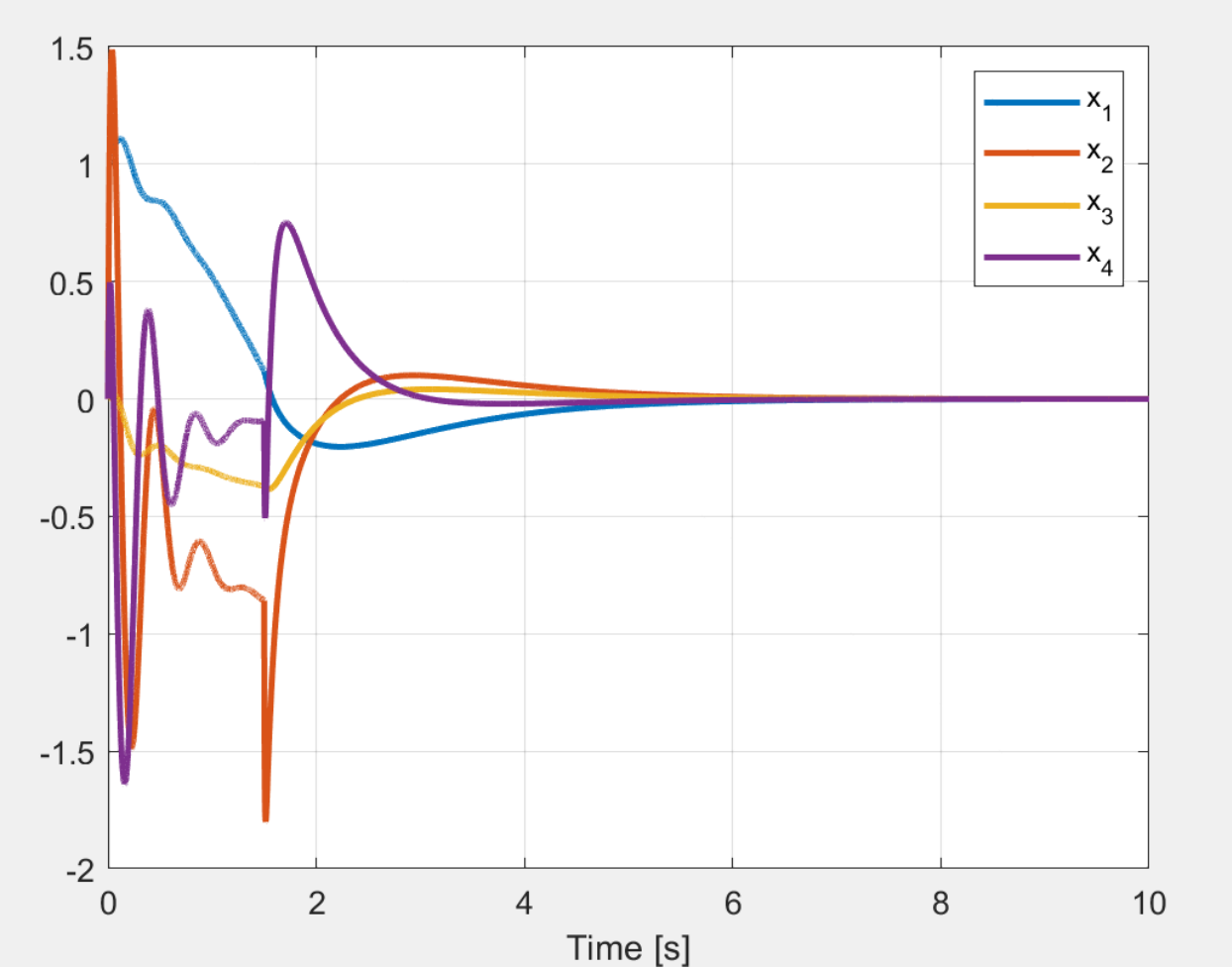


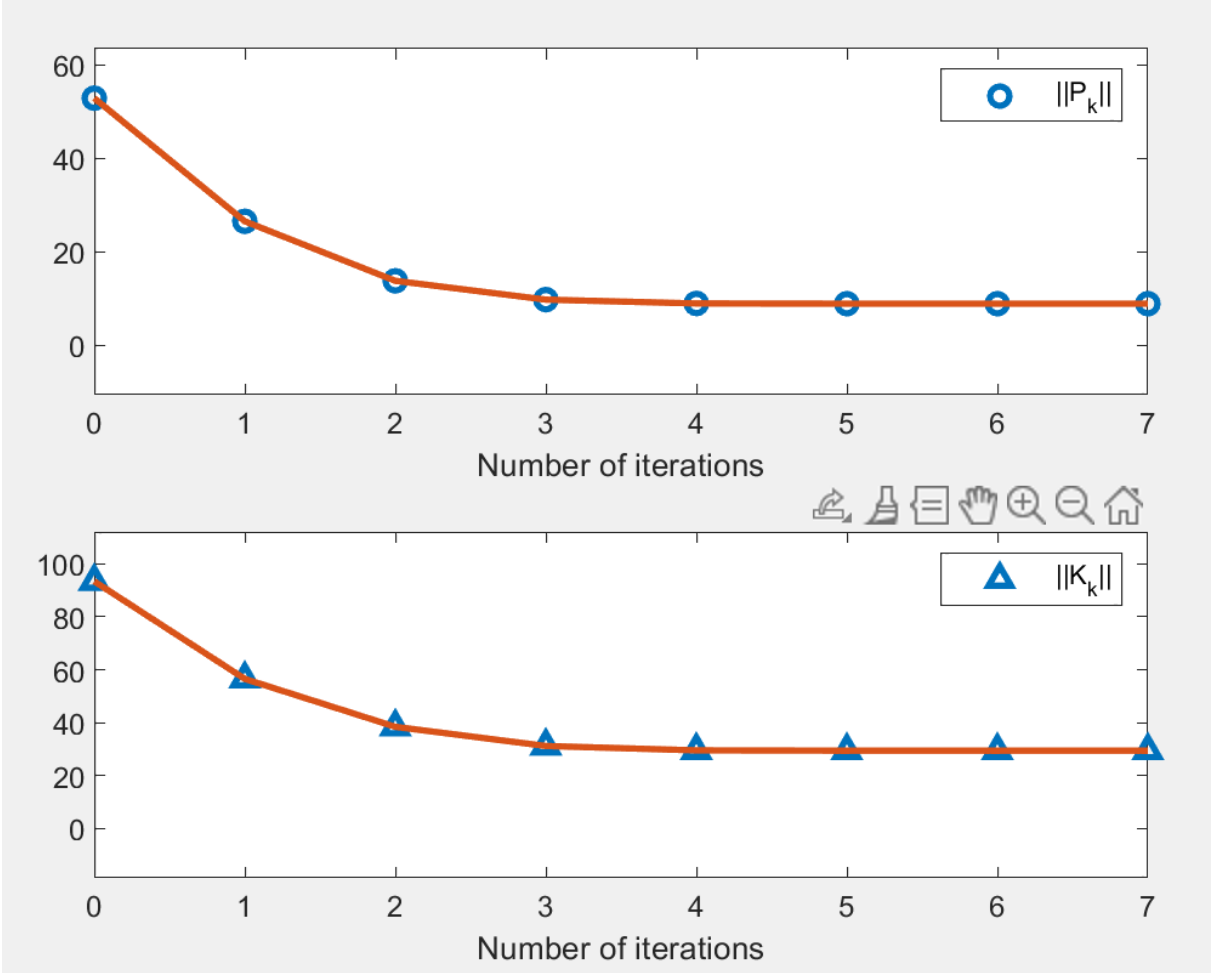
$$Q = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R = 0.1$$

$$K = [-3.16 \quad 7.12 \quad -27.69 \quad -6.11]$$

Data-Driven ADP for LQR with unknown (A, B)





$$K^* = [-3.16 \quad 7.12 \quad -27.69 \quad -6.11]$$

References

- Sveen, Emil Mühlbradt & Zhou, Jing (2023). Applications of Linear Adaptive Dynamic Programming (ADP) to a Non-linear Four-bar Mechanism. *International Conference on Control, Mechatronics and Automation.* 2023

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